



Synchronisierung von ArchiMate und BPMN Modellen: Eine Literaturübersicht

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Erdoğan Yıldız

Matrikelnummer 09925759

an der Fakultät für Informatik

der Technischen Universität Wien

Betreuung: Univ. Prof. Henderik Proper, PhD

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Erdoğan Yıldız

Henderik Proper



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Erdoğan Yıldız

Registration Number 09925759

to the Faculty of Informatics

at the TU Wien

Advisor: Univ. Prof. Henderik Proper, PhD

Vienna, February 1, 2025

Erdoğan Yıldız

Henderik Proper

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Kurzfassung

Die Modellierungssprache ArchiMate hat sich als de-facto Standard für die Visualisierung von Unternehmensarchitekturen etabliert. Sie ermöglicht die Darstellung von Geschäftsprozessen, Applikationen und Technologien und kann durch Erweiterungen sogar Strategien, Implementierungen und Migrationen abbilden. Neben diesen drei zentralen Ebenen erlaubt ArchiMate auch die Modellierung von Abhängigkeiten zwischen den Abstraktionsebenen, wodurch eine höhere Transparenz innerhalb des Unternehmens geschaffen wird. Die Business Process Model and Notation (BPMN), entwickelt von The Object Management Group (OMG), gilt in vielen Unternehmen als Standard für die detaillierte Modellierung von Geschäftsprozessen. Während ArchiMate Geschäftsprozesse auf einer abstrakteren Ebene darstellt, bietet BPMN eine präzisere und detailliertere Abbildung dieser Prozesse. Aufgrund dieser unterschiedlichen Detaillierungsgrade besteht zwischen beiden Modellierungssprachen eine inhaltliche Verbindung.

Die zentrale Herausforderung besteht darin, eine Synchronisation von Modellen beider Sprachen herzustellen – insbesondere von der Geschäftsebene in ArchiMate hin zu detaillierten Prozessmodellen in BPMN und zurück.

Diese Arbeit untersucht anhand einer kritischen Literaturrecherche beide Modellierungssprachen im Detail und analysiert verschiedene Konzepte zur Synchronisation. Zudem werden Ansätze zur Filterung von Informationen bei der Überführung von BPMN-Modellen in ArchiMate betrachtet. Dies kann dazu beitragen, die Datenflut zu kontrollieren und gleichzeitig die Flexibilität in der Modellierung zu bewahren. Eine steuerbare Synchronisation zwischen ArchiMate und BPMN ermöglicht es, detaillierte Geschäftsprozesse aus abstrakteren Unternehmensarchitekturen abzuleiten und umgekehrt bestehende BPMN-Modelle in eine höhere Abstraktionsebene zu überführen. Dadurch kann diese Synchronisation Unternehmen dabei unterstützen, bessere strategische Entscheidungen zu treffen.

Abstract

The ArchiMate modeling language has established itself as the defacto standard for the representation of enterprise architectures. It enables the representation of business processes, applications and technologies and can even capture strategies, as well as implementation and migration plans through its language extensions. In addition to these three central levels, ArchiMate also allows for the modeling of dependencies between different levels of abstraction, creating greater transparency within the organisation.

The Business Process Model and Notation (BPMN), developed by the Object Management Group (OMG), is considered the standard for detailed modeling of business processes in many companies. While ArchiMate's business layer represents business processes at a more abstract level, BPMN offers a more precise and detailed representation of these processes. Due to these different levels of detail, but shared focus,, there is a connection between the two modeling languages.

The key challenge is to *synchronize* models from both languages – in particular, from the business level in ArchiMate to detailed process models in BPMN and back.

This thesis examines both modeling languages in detail based on a critical literature review, while also analyzing various synchronization concepts. A key focus when analyzing synchronization approaches is on the filtering of concepts between the two standards. This is particularly important when synchronizing detailed BPMN models to the coarse granular ArchiMate models, since many BPMN concepts cannot be directly mapped in ArchiMate.

Controlled synchronization between ArchiMate and BPMN makes it possible to relate detailed business processes to more abstract enterprise architectures. This synchronization can thus, help companies to make better strategic decisions. The extent to which BPMN models can be (partially) derived automatically from ArchiMate (business layer) models, and vice versa, will be touched upon as well in this thesis.

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Introduction

Although enterprise architecture (EA) represents a coherent set of principles, methods and models for designing organisational structure, business processes and information systems [Lan17], different organisational units within an organisation often develop their own models. Even more, it is likely that different techniques and languages are used to express these models, resulting in a multitude of architecture descriptions. This heterogeneity complicates both communication and decision-making within the organisation [vLtDL04]. The link between ArchiMate models and Business Process Model and Notation (BPMN) models has been reported on in different papers ([Lan17, Pen13, Gil15]). In these papers, the integration of models and concepts from enterprise architecture with specialized modeling tools such as BPMN is often described as a bidirectional link [Lan17]. The approach taken includes the extraction of enterprise architecture artifacts, taking into account existing architecture frameworks and languages [OK20]. ArchiMate provides a high-level enterprise modeling, while BPMN enables much more detailed process modeling.

The relationship between the two languages can also be viewed as a T-shaped model. A T-shaped model is used in various areas of management, for example in skill management, knowledge management or project management. The principle of this model in the context of ArchiMate and BPMN is shown in Figure 1.1. The horizontal axis represents the layers of the enterprise architecture, while the vertical axis represents the level of abstraction at which more detailed models can be created. In principle, vertical detailing can be done from any layer of the enterprise architecture. However, the focus of this work is not on a comprehensive consideration of all layers, but specifically on the synchronization between ArchiMate and BPMN.

While the research by [OK20] investigated the creation of ArchiMate models from BPMN models, so-called the “bottom-up” approach, [Gil15] research analyzes the applicability of extracting BPMN models from ArchiMate diagrams, the so-called “top-down” approach. This Bachelor thesis investigates the extent to which the de facto standard ArchiMate for enterprise architecture can be combined with more detailed and specific modeling

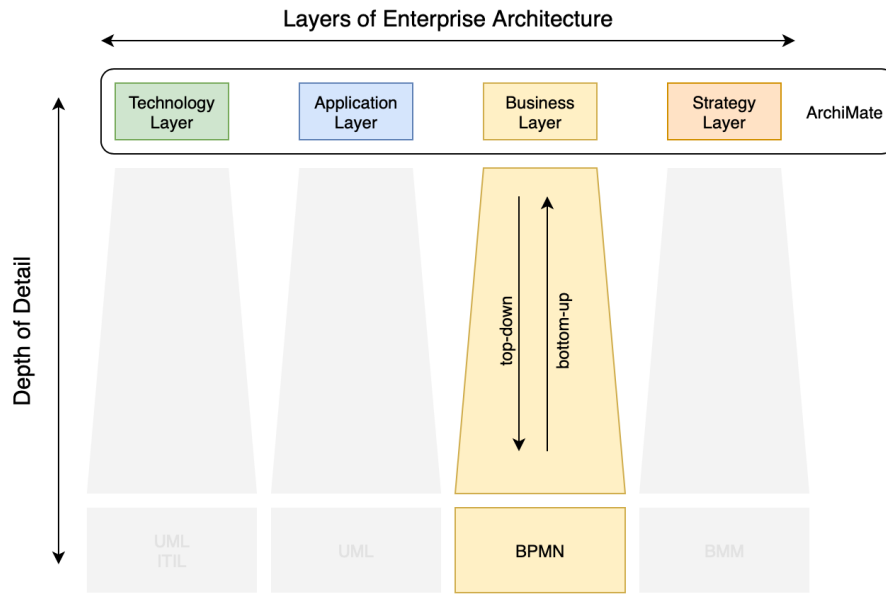


Figure 1.1: T-shaped model of the relationship between ArchiMate and more detailed modeling standards.

standards such as BPMN. In [Pen13], an evaluation was also carried out using the Bunge-Wand-Weber (BWW) model [WW93], which provided insights into the completeness and semantic consistency of business process models.

The basis for the “top-down” and “bottom-up” approach to linking abstract ArchiMate models with detailed models of different domains was described in [vLtDL04] as a “workbench”. This concept is based on the well-known Model-View-Controller (MVC) pattern [Buc09] from software engineering.

Since the BPMN modeling language offers a greater range of concepts and elements for business process modeling than ArchiMate, it is obvious that when synchronizing from BPMN to ArchiMate, certain BPMN concepts cannot be depicted in ArchiMate. To ensure a meaningful and consistent transfer nonetheless, a controlled synchronization in the form of filtering is necessary, as otherwise semantically incorrect models could result.

The current state of research and the existing research gap in model synchronization lead to the following research questions: “*How can ArchiMate and BPMN be linked in such a way that synchronization across different levels of abstraction is ensured without semantic deviations?*” and “*What explicit filter mechanisms do these synchronization strategies provide when generating ArchiMate models from BPMN models?*”

The goal of this thesis is to examine the synchronization of ArchiMate and BPMN models by analyzing the top-down and bottom-up approaches. While the top-down approach enables the conversion of abstract enterprise models into detailed process diagrams, the bottom-up approach facilitates the integration of existing business processes into

enterprise architecture diagrams. The objective is to systematically investigate how both approaches can be applied, what challenges arise in each case, which synchronization approach has an explicit filter, and how a strategy for filtering could look like. A critical literature review is to be conducted to identify the research gaps and missing studies in this area, in order to provide targeted support for future research.

At the start of this thesis, the theoretical foundations of the two standards ArchiMate of The Open Group and Business Process Model and Notation (BPMN) of the Object Management Group (OMG) are explained in more detail in chapter 2. Chapter 3 provides an overview of existing synchronization approaches, which are compared with each other. Finally, chapter 4 summarizes the literature research and addresses outstanding research questions.

Foundations

This chapter explains the basics of ArchiMate from The Open Group and Business Process Model and Notation (BPMN) from the Object Management Group. It summarizes key terms, definitions, metamodels, elements, and the relationships between these elements. The aim is to provide a basic understanding of both modeling languages.

2.1 ArchiMate 3.2

The role of an architect at any level of an organisation involves taking on the concerns and requirements of various stakeholders, analyzing them, and implementing them in an enterprise architecture [TOG22].

The ArchiMate is a standard that is managed by The Open Group, and involves a modeling language to visually represent enterprise architecture concerns and requirements using a standardized set of signs and symbols. The standard makes it possible to represent concerns and requirements using entities and their relationships to promote a common understanding of requirements and problems at different levels of the organisation. In addition, the ArchiMate standard also includes an XML-based representation format, to facilitate the exchange of models and data between different applications [TOG22].

ArchiMate provides a unified notation for diagrams, including concepts for documenting interconnected architectures, specific perspectives, and mechanisms for adapting the language. The integrated architecture approach supports the description and visualization of different architecture domains and their relationships and dependencies. Notation and model elements are distinguished from each other in order to clearly separate the presentation of stakeholder needs from architecture information. The language is service-oriented and makes it possible to represent and link the business, application and technology layers of an organisation separately. These three layers can also be integrated across different levels [TOG22].

2.1.1 Core Principles

General metamodels for enterprise architectures face the challenge of finding a balance between languages for describing specific architecture domains and general architecture concepts for representing system perspectives. The ArchiMate language is based on relatively general concepts that have been specifically adapted for the different architecture levels. It was deliberately kept as small as possible to maintain the simplicity of the language, but is still sufficiently suitable to cover most enterprise architecture modeling tasks. Although this simplicity does impose certain restrictions, it is still perfectly adequate for about 80% of practical use cases [TOG22].

A more detailed and accurate description and visual representation of requirements, for example, for business processes, can be created using BPMN 2.0. The above-mentioned limitations and the fact that ArchiMate has limited use for detailed process descriptions raises the question of how BPMN models can be created from ArchiMate models. In the opposite direction, the question arises as to how an ArchiMate model can be created from a BPMN model if ArchiMate offers restrictions in the language.

2.1.2 Layering of the ArchiMate Language

As shown in Figure 2.1, the ArchiMate core language is divided into three layers, and the generic elements and relationships are specified in these layers.

- The *Business Layer* represents the organisation's business processes, which are carried out by business actors.
- The *Application Layer* represents the services that support the business and the applications that implement them.
- The *Technology Layer* is used to represent the information and operating technology in the company. This layer can be used to model the operational or physical technologies that support the applications and business processes.

These three layers represent the core framework and essentially represent the minimum required to model architectural models. The standard extends the technology layer by a physical layer in the complete framework. In addition, an extra layer, implementation and migration, is introduced, which is specifically intended for modeling implementations and migrations. The enterprise strategies and choices are mapped in the strategy layer. The motivation aspect is taken into account across all layers. The layers are also divided into the aspects Passive Structure, Behavior and Active Structure. The Active Structure aspect represents structural elements such as business actors or application components. The Behavior aspect represents the behavior performed by the actors. The passive structure aspect describes objects that are performed by the behavior. The special thing about this aspects is that it is based on the natural structure of language with

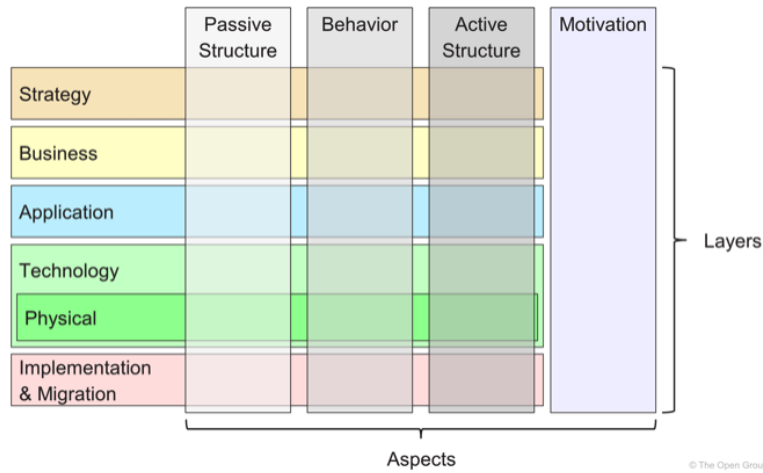


Figure 2.1: Layering of the ArchiMate Language. Adpoted from [TOG22]

subject (active structure), verb (behavior) and object (passive structure), which makes the modeling easy to understand [TOG22].

Since our focus is on the business layer, the complete framework, as shown in Figure 2.1, is not relevant for this thesis and will therefore not be discussed further.

2.1.3 High Level Language Structure

Figure 2.3 shows the abstract high-level hierarchy of the ArchiMate concept. A *Model* is a collection of *Concepts* and represents the highest element. *Element*, *Relationship* and *Relationship Connector* are specializations of *Concept*. An *Element* is either a *Behavior*, *Structure*, *Motivation* or *Composite*. ArchiMate provides a basic set of generic relationships shown in Figure 2.2 that can be used to link source and target concepts. The relationships are divided into four different classes.

- *Structural Relationship* are used to model static constructions or the composition of similar or different concepts.
- *Dependency relationship* are used to model elements that support other elements.
- *Dynamic relationship* describes the behavior between elements.
- *Other relationships* include all relationships that do not fall into the above classifications.

The Element is divided into two main types.

- *Behavior Elements* are subdivided into External Behavior Elements (also called services), Internal Behavior Elements and Events.

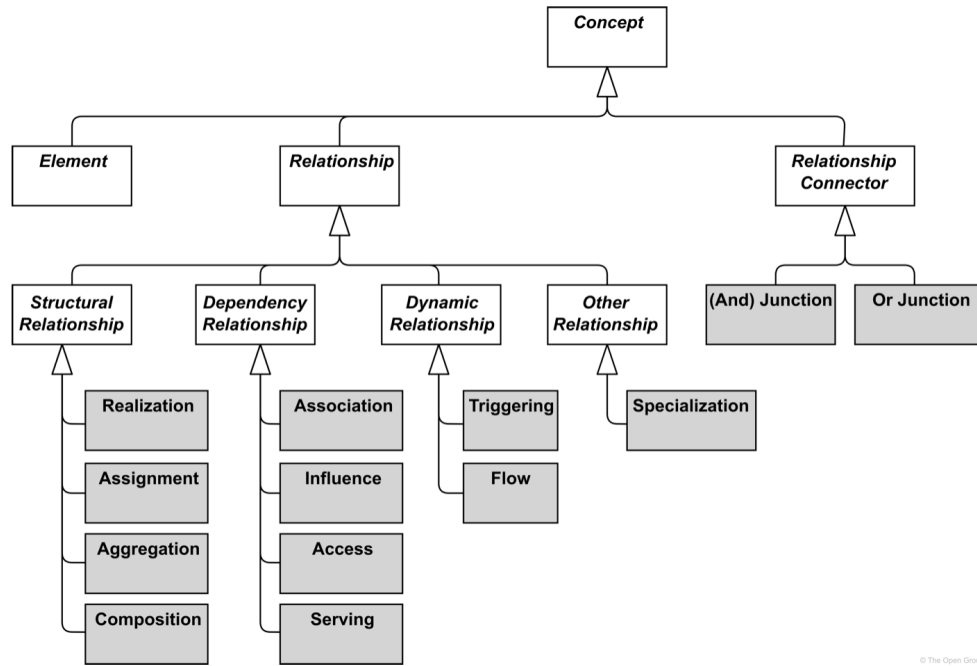


Figure 2.2: ArchiMate Relationship types. Adopted from [TOG22]

- *Structure Elements* are differentiated into Active Structure Element and Passive Structure Elements. The Active Structure Element are further subdivided into External Active Structure Element and Internal Active Structure Element.

External Elements are also called Interfaces and represent external access points [TOG22].

This summary contains the most important information about the abstract, generic meta-model. The elements described are layer-independent and are not used directly in the models, but only serve to describe the concepts. Specializations of these elements are derived in the individual layers. An overview of the more specific elements is shown in Table 2.1 and relationships of the business layer is shown in Table 2.1. A comprehensive and detailed description can be found in the ArchiMate Specification [TOG22].

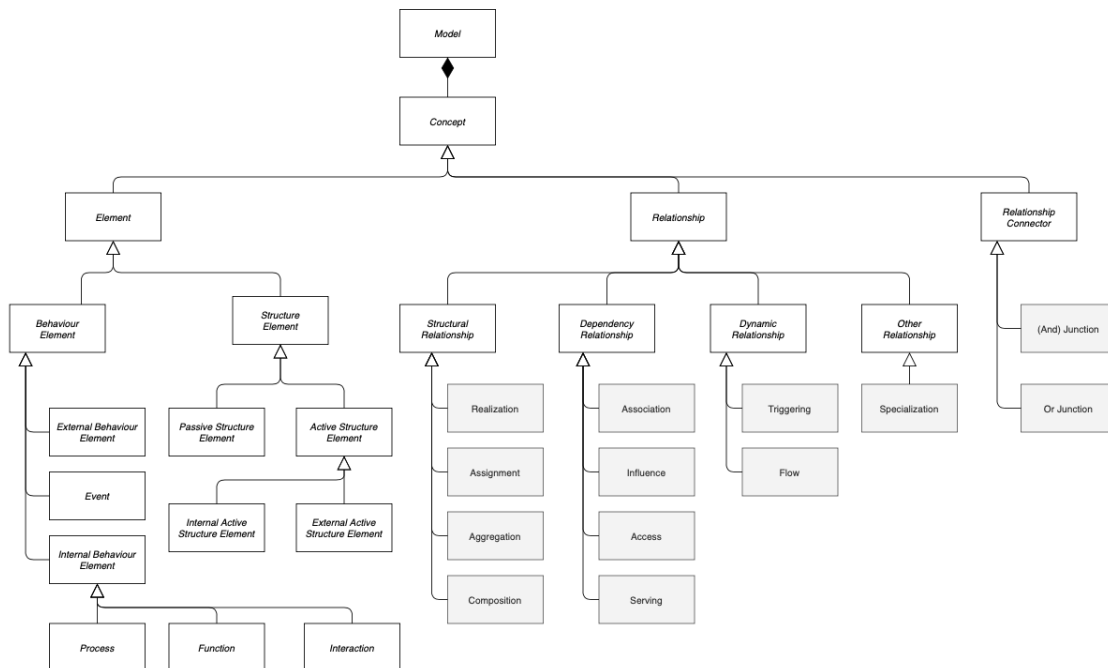


Figure 2.3: ArchiMate High-Level Structure. Adopted from [TOG22].

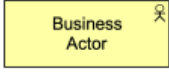
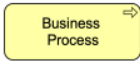
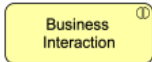
Active Structure Elements		
Element	Description	Notation
Business Actor	A business unit that can organize an event.	 
Business Role	Responsible actor who triggers a certain behavior.	 
Business Collaboration	Collaboration between two or more business-related active elements that jointly execute a behavior.	 
Business Interface	The business service available to the environment.	 
Behavior Elements		
Element	Description	Notation
Business Process	Is a sequence of business behaviors with the aim of achieving a result.	 
Business Function	A collection of business behavior that is managed as a whole based on a selected set of criteria.	 
Business Interaction	A collective unit that is performed by two or more business actors or business roles.	 

Table 2.1: Part of the Active Structure Elements and Behavior Elements from the Business Layer. Adopted from [TOG22]

2.2 Business Process Model and Notation (BPMN)

A central task of business people and business analysts is to analyze and document business processes in order to create transparency and at the same time identify optimization potential, which is an essential part of a company's economic orientation.

In recent decades, web service-based XML languages have been continuously developed to provide formal mechanisms for executing business processes. Although optimized for automated execution in software systems, they provide limited capabilities for human users in terms of interoperability and portability. The Business Process Model and Notation (BPMN), introduced in version 2.0 in 2011, was developed by the Object Management Group (OMG), a consortium of companies, government organisations, and research institutions. The goal of this notation was to bridge the gap between human comprehensibility and machine-readable language by providing an intuitive modeling language. BPMN also enables the visual representation of web service-based XML languages [vRWCdM15].

2.2.1 Models and Elements

Business Process Modeling (BPM) is designed to communicate a large amount of information to a broad audience in a comprehensible way. To meet this requirement, BPMN divides the models into the following sub-models [vRWCdM15]:

Processes (Orchestration) are a sequence of activities with the aim of fulfilling certain tasks. These can be subdivided into *Private (internal) Business Processes* and *Public Processes*. Private Business processes represent the internal orchestration of services. They serve exclusively internal purposes and are not visible to external parties. A process is typically represented in a pool, while communication between different pools is visualized using message flows. Figure 2.4 shows a typical example of a public and a private process [vRWCdM15].

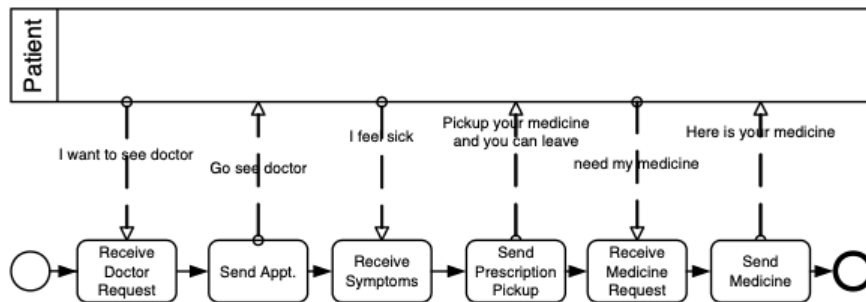


Figure 2.4: Example of a *private* and *public* Business Process. Adopted from [vRWCdM15].

A **Collaboration** represents the interaction between two or more business units organized in different pools. In comparison to processes, where the communication via message

flows ends at the border of a pool, in a collaboration the activities are directly linked with each other, thus visualizing the interaction more clearly. In this sub-model, all combinations of pools, processes and choreographies are allowed. Figure 2.5 shows an example of a collaboration [vRWCdM15].

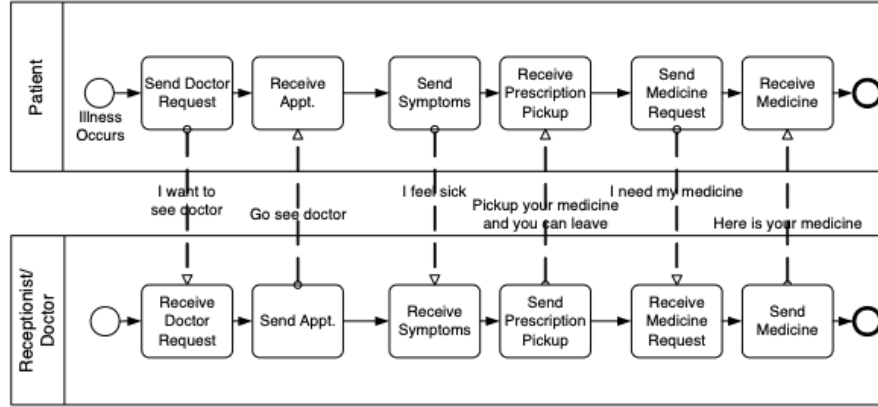


Figure 2.5: An example of a Collaborative Process. Adopted from [vRWCdM15].

Choreographies are independent models and represent a special type of process. Similar to private business processes, they display workflows, but with a focus on interactions and message exchange between participants. In contrary to processes, there is no central instance responsible or monitoring in a choreography. Rather, it functions more like a procedural contract between the participants. Figure 2.6 shows an example of a choreography [vRWCdM15].

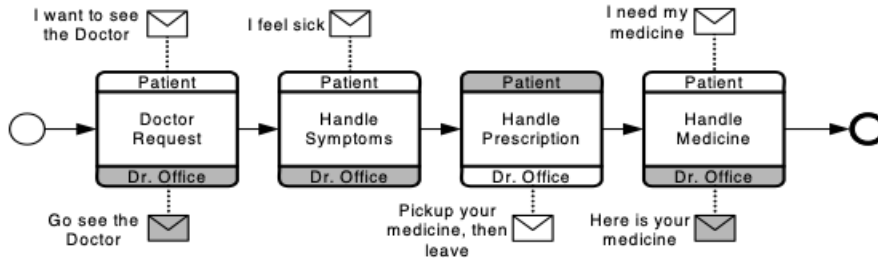


Figure 2.6: An example of a Choreography. Adopted from [vRWCdM15].

The graphical elements in BPMN are organized into five categories, which makes them easier for users to recognize and understand. Although adjustments are possible within these categories depending on the requirements, the basic appearance remains largely unchanged [vRWCdM15]. An overview of the categories and their elements is shown in Table 2.2.

Categories		
Category	Element	Notation
Flow Objects	<i>Main behavior objects of a Business Process.</i>	
	Event	Something that happens
	Activity	A term for work
	Gateway	Control flow decisions
Data	<i>Provide information for processes.</i>	
	Data Objects	Represent information
	Data Inputs	Required data
	Data Outputs	Produced data
	Data Stores	Persistent data
Connecting Objects	<i>Define relationships between elements.</i>	
	Sequence Flows	Order of activities
	Message Flows	Flow of messages
	Associations	Link information and Artifacts
	Data Associations	Data linkage
Swimlanes	<i>Organize Participants in a Collaboration</i>	
	Pools	Major process participants
	Lanes	Functional responsibilities
Artifacts	<i>Provide additional information.</i>	
	Group	Logical grouping
	Text Annotation	Additional notes

Table 2.2: A summary of categories and elements in BPMN 2.0. Adapted from [vRWCdM15].

The fundamentals of ArchiMate 3.2 and BPMN 2.0 shown here illustrate the differences in the application areas of both modeling languages as well as their respective boundaries within their domains. This summary also highlights their differences and similarities – in particular the fact that both languages model business processes, but at different levels of abstraction. As described in the introduction, the challenge is to synchronize models between ArchiMate and BPMN across these levels of abstraction, taking into account possible filtering mechanisms. The following chapter 3 examines how synchronization between the models can be implemented. To do this, a critical analysis of existing approaches is carried out with regard to their applicability and challenges. This analysis is carried out as part of a critical literature review, in which relevant scientific work is systematically examined, evaluated and compared.

Existing Synchronization Approaches

3.1 Methodology

In order to answer the research question of this bachelor thesis, a critical literature review was conducted. This research provided qualitative results that help to answer the research question. The following databases and search platforms were used to search for relevant scientific papers:

- Google Scholar
- IEEE Xplore
- Springer Link

The search strategy included a combination of the following terms: ArchiMate, BPMN, Linkage, Synchronization, Enterprise Architecture and related keywords such as:

- "archimate and bpmn integration"
- "bpmn and archimate synchronization"
- "linkage bpmn and archimate"
- "synchronization of archimate and bpmn"
- "Architecting business process maps"

Nr	Criteria	Description	Attribute
1	Reference	Reference in the bibliography.	Citation
2	Author(s)	Author(s) of the work.	Name of the author(s)
3	Typ of link	Direct Establishes a direct link between elements from heterogeneous models. New meta model Uses an additional meta model as a synchronization model to link heterogeneous models.	Direct/New meta model
4	Year of publication	The year of publication of the respective scientific work, in order to be able to classify the temporal context and the state of research.	<YYYY>
5	Synchronization direction	Top-down generation of detailed business process models from enterprise architecture models. Bottom-up derivation of enterprise architecture models from detailed business process models.	top-down / bottom-up
6	Proven	Indicates whether the approach has been validated by practical examples or in other scientific papers.	yes/no

Table 3.1: Criteria attributes for the literature review.

The selection of the final literature was based on a critical review of the relevance of the abstract, citation frequency, indirect references in other works.

In this chapter, the identified existing approaches are summarized and critically analyzed. The aim is to examine where research gaps exist and in which areas well-founded work has already been done. At the end of this chapter, the various approaches are compared in a table based on defined criteria. The criteria for the tabular comparison of the approaches are defined in Table 3.1.

In this qualitative research, transparency is ensured by the fact that all work steps are comprehensively documented. Each approach and each cited literature is considered subjectively and openly put up for discussion.

3.2 Related works

The literature research has shown that the linking and synchronization of these two standards presents complex challenges and has also been little researched so far.

According to Lankhorst [Lan17], modeling languages such as BPMN do not capture all

aspects of an “overall” language. A key problem is that the relationships between the domains (views) are insufficiently defined and difficult to link. Furthermore, they lack a formal basis and clearly defined semantics, which means that they cannot represent comprehensive architectural visions. Penicina highlights the challenge of “completeness of process models” in [Pen13] and emphasizes that, from an information system perspective, models must contain all necessary elements must be included. As described in the previous chapter, the two standards differ in their scope. Gill [Gil15] also points out that a top-down approach is complicated by the different levels of detail in the modeling standards – from high-level to detailed low-level models. Another significant challenge is described in [SSA24] from the perspective of the user. Sanyoto emphasizes that many companies already use comprehensive architecture descriptions and models, which makes a complete switch to a new modeling language impractical. This underlines the relevance of synchronization between existing standards. Gonzales-Lopez and Bustos [GLB19] point out the technological developments of recent decades and emphasize that business processes increasingly extend beyond the boundaries of individual organisations. Their literature review shows that the research focus is either on enterprise architecture or on business process architecture, while an integrative view of both areas is largely missing.

3.3 Lankhorst et al. (2004) approach to model integration

An important approach to integrating ArchiMate with other modeling languages is described by Lankhorst et al. as a “Workbench”. This approach is based on the concept of viewpoints, which is explained in detail in [TOG22], and forms the basis for implementing the integration. In doing so, various design principles are followed, with one of the central principles being the seamless integration of existing modeling schemas. In addition, the approach is extensible, so that new modeling standards can be flexibly integrated. The ArchiMate “is-a” relationship approach is used to minimize the loss of information in bottom-up and top-down approaches. [vLtDL04].

Figure 3.1 illustrates the architecture of Lankhorst et al.’s Workbench idea, which is divided into three layers. The first layer, the *Workbench tier*, contains the *ArchiMate workbench* as a central element. Only models that are defined in the *ArchiMate model* and meet the requirements of the *ArchiMate Viewpoint* are allowed here. In the third layer, the **Tool tier**, specific modeling tools are integrated. Similar to the *Workbench tier*, this is done with the help of a *Modeling schema* and a *Tool-specific model*. The *Integration tier* is responsible for the central part between these two layers, in particular through the *Tool-specific adapter* [vLtDL04].

Lankhorst describes in [vLtDL04] two central approaches to the integration of heterogeneous models. These approaches will be examined in more detail in the further course of this chapter 3 in the critical analysis of the methods identified in the literature research and used as comparison attributes.

- **Direct model integration** A direct link between concepts of heterogeneous

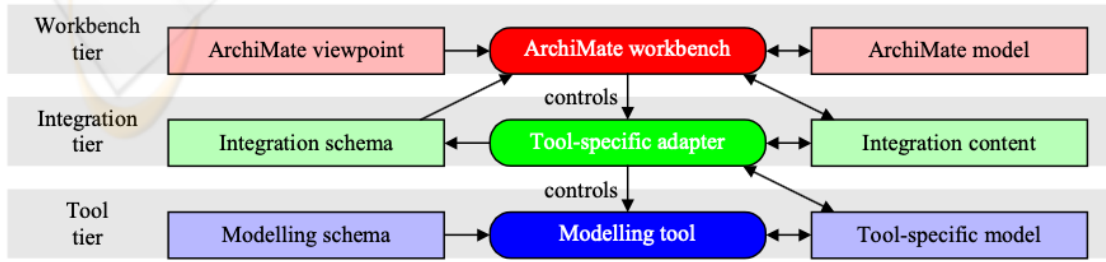


Figure 3.1: The 3-tier workbench architecture. Adopted from [vLtDL04]

languages using an ontology. The disadvantage of this approach lies in the number of links, which scales with $O(n)$ to the number of other languages n .

- **Integrated metamodels** A newly developed meta-model that serves as a mapping model for heterogeneous languages.

This approach to integration offers a promising abstract solution for linking heterogeneous models using an integrated meta-model. However, the concrete implementation within the so-called “tool-specific adapter” is missing in this work, which would be essential for practical application. In addition, the viewpoints approach only allows a partial representation of a model, which means that only a limited view of the overall system is possible. In large, process-oriented companies, this solution could therefore pose challenges rather than offering practical benefits.

Although the concept of integration schemes and integration content is not covered in detail in this paper, it does provide approaches to how content could be filtered using the bottom-up approach. The approach describes how the Workbench and the adapter first generate integration content from the underlying specific model and then create an integration schema based on this content. Filtering during synchronization could be integrated into this process [vLtDL04].

3.4 Orlovskiy and Kopp (2020) data extraction approach from business process models

Orlovskiy and Kopp present a direct mapping approach for heterogeneous models in [OK20]. They explain how ArchiMate models can be extracted from existing BPMN models using a bottom-up approach. They point out that previous research has shown that a direct mapping between ArchiMate and BPMN elements often has many-to-many relationships, which leads to uncertainty. To mitigate this problem, they prefer a 1-to-1 mapping in their work to ensure a clear and unambiguous mapping of the models. Figure 3.3 shows the conceptual scheme of a possible implementation [OK20].

They recommend converting the meta-models into a collection of RDF (Resource Description Framework) statements, which are represented in the form of “subject-predicate-

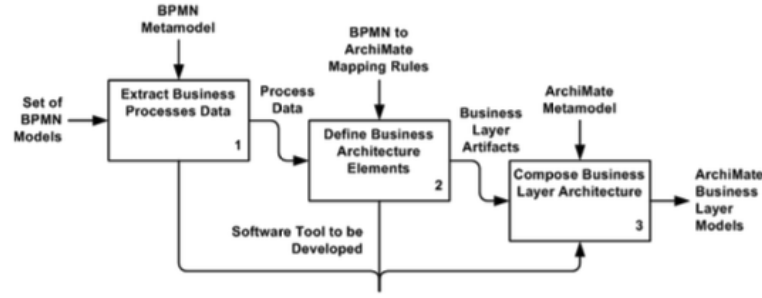


Figure 3.2: Conceptual scheme of the considered problem. Adopted from [OK20].

object” tuples. This results in RDF graphs that can be used to define mapping rules. As a query language, they suggest SPARQL because this method makes it possible to query specifically which incoming or outgoing links an element can have. The following examples show how such queries can look and what results they provide [OK20].

```
SELECT * WHERE { <Event> ?a ?x . }
SELECT * WHERE { <Business Event> ?b ?y . }
```

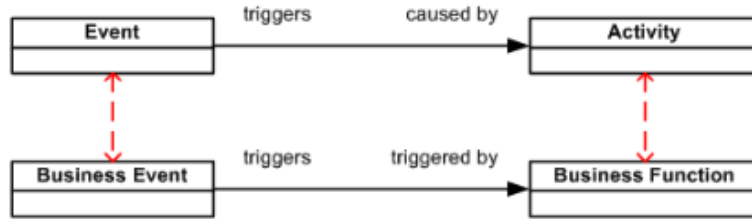


Figure 3.3: Results of these SPARQL query. Adopted from [OK20].

Interoperability was achieved by modifying a breadth-first search algorithm that could create a one-to-one mapping. This made it possible to generate a valid mapping table between ArchiMate and BPMN elements. The results, shown in Figure 3.4, of the breadth-first search algorithm resulted in unique one-to-one relationships that were used to validate the manual queries [OK20].

Orlovskiy and Kopp also propose a software solution based on this approach. They recommend using the XML standard for information exchange. They support their approach with concrete examples and show that it enables valid extractions from BPMN models. They also prove that the maintenance effort is kept low by calculating using the propagation cost measure method [OK20].

The results of Orlovskiy et al. on the 1-to-1 link table enable an analysis of the interface between the ArchiMate and BPMN modeling languages. This table represents a one-to-one mapping. Nevertheless, there are challenges in the practical application of this direct link.

Node	Statements (BPMN)	Node	Statements (ArchiMate)
Event	Event triggers Activity	Event	Event triggers Function
Activity	Activity uses Data Object, Activity sequences Activity, Gateway controls Activity, Pool contains Activity	Function	Function accesses Object, Function triggers Function, Process aggregates Function
Pool	Lane sub-divides Pool, Message flows between Pool	Process	Process realizes Service, Role assigned to Process

Figure 3.4: Results of BPMN and ArchiMate meta-models traversing. Adopted from [OK20].

One problem arises when mapping BPMN pools to the ArchiMate Business Process element. For example, in Figure 3.4, the relationship *Lane sub-divides Pool* is interpreted as “Process realizes Service”.

According to this definition, a pool typically represents a business unit or even an entire organisation. This can lead to problems when translating BPMN models into ArchiMate models, as the original meaning may not be correctly preserved.

On the other hand, this work focuses exclusively on the analysis of a direct mapping and the bottom-up approach. Although the resulting 1-to-1 mapping of concepts provides a clear mapping, it does not contain a strategy for implementing filter mechanisms. Nevertheless, the findings provide scientific value and can serve as a basis for future research on synchronization.

3.5 Linkage of ArchiMate and BPMN with the Bunge-Wand-Weber model

The approach of the author of [Pen13] is to check ArchiMate and BPMN models for completeness and semantic correctness using the theoretical framework of the Bunge-Wand-Weber (BWW) model. The BWW model, a high-level ontology used to model information systems, is intended to act as an “intermediary” between the ArchiMate and BPMN standards. To make this possible, the author suggests using a repository-based modeling tool that supports all three methods – ArchiMate, BPMN and BWW.

In an earlier paper by Penicina et al. [PK13], in addition to the integration of ArchiMate, BPMN and BWW, the connection of external, maintained regulations is also recommended. This external link should help to ensure the completeness and semantic correctness of the models. The high-level architecture proposed by the authors is visualized in Figure 3.5.

As suggested in [TOG22], detailed business processes are extended by business process modeling. In line with this, the author in [Pen13], tries to link ArchiMate business layer elements with BPMN elements in the first step. The result of these many-to-many links can be seen in Figure 3.6.

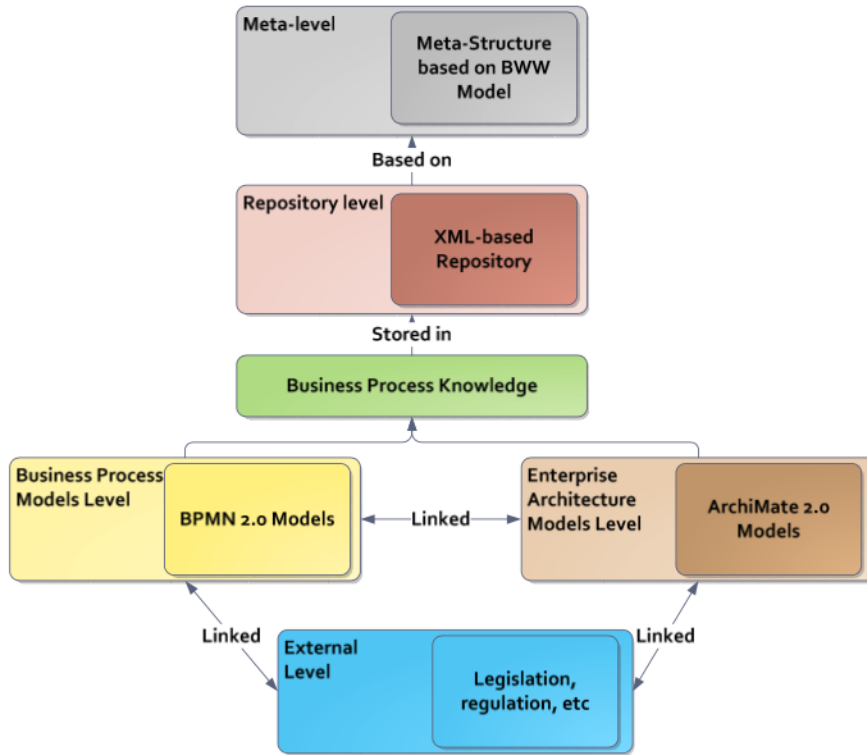


Figure 3.5: The high-level architecture between BPMN, ArchiMate and regulations. Adopted from [PK13].

In the next step, the BWW elements are linked to the corresponding ArchiMate elements. It is particularly important that the general enterprise architecture is integrated as a system and the layers as subsystems from the BWW model. The BWW State element is represented by BPMN Data Objects and various End Events of the process. However, not all BWW elements can be linked to ArchiMate and BPMN. Specifically, 11 BWW elements are incompatible with ArchiMate and 6 BWW elements are incompatible with BPMN. A detailed mapping of these model is shown in Figure 3.7 [Pen13].

This approach of combining ArchiMate and BPMN with the BWW model has great potential and certainly requires further research. It also enables the integration of external regulations and contributes to the completeness and semantic correctness of model mappings. However, the approach remains incomplete due to the lack of support for 11 BWW elements in ArchiMate and 6 BWW elements in BPMN, which indicates existing research gaps.

This paper proposes a business process knowledge layer based on a meta-model of the Bunge-Wand-Weber model (BWW). This layer connects the ArchiMate and BPMN models as well as external regulations to ensure semantically correct storage. At the same time, this business process knowledge approach would provide the option of implementing

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ArchiMate Business Layer Element	BPMN Element
Business Process	Business Process Diagram, Pools, Lanes
Function	Task, Sub-Process
Business Interaction	Collaboration Diagram
Business Event	Event
Business Object	Data Object
Business Role	Lane
ArchiMate Application Layer Element	BPMN Element
Application Function	Service Task, Script Task
Data Object	Data Object
ArchiMate Technology Layer Element	BPMN Element
Device	Data Store
Artefact	Data Objects

Figure 3.6: Mapping ArchiMate and BPMN. Adopted from [Pen13]

controlled filtering when generating ArchiMate models from BPMN models.

BWW Elements	Corresponding ArchiMate Concepts
System	Enterprise architecture, Business layer, Application layer, Technology layer
System environment	Business layer (for Application and Technology layers), Application layer (for Technology and Business layers), Technology layer (for Business and Application layers)
System structure	Relationships
System composition	Structural concepts, Informational concepts
Level structure	Relationships between layers, Business layer, Application layer, Technology layer
Subsystem	Business layer, Application layer, Technology layer
System decomposition	Business layer, Application layer, Technology layer, Structural concepts, Informational concepts
Thing	Business actor, Business role, Business collaboration, Location, Business interface, Business object, Application component, Application collaboration, Application interface, Data object, Node, Device, System software, Infrastructure interface, Network, Communication path, Product, Contract, Artifact
Property	Meaning, Value, Representation
Class, Kind	Relationships
Event, External event, Internal event, Poorly-defined event	Business event
Transformation	Business service, Business process, Business function, Business interaction, Application function, Application interaction, Application service, Infrastructure function, Infrastructure service
Acts on, Coupling	Structural relationships
State, Conceivable state space, Lawful state space, State law, Stable state, Unstable state, History, Conceivable event space, Lawful event space, Well-defined event, Lawful transformation	Not supported by ArchiMate

Figure 3.7: BWW elements mapped into ArchiMate elements. Adopted from [Pen13]

To sum up, the link between ArchiMate and BPMN is indirect via the BWW model. While ArchiMate represents the structural and behavioral aspects of an organisation, BPMN remains responsible for process modeling. The data objects and final events in BPMN, which are supposed to represent the state, express the state of a system and thus form the link between the models. Although the BWW model can be used to check the completeness of models, the approach remains incomplete due to the lack of links to the BWW elements and requires further research [Pen13].

3.6 Applicability and Interoperability Research by Gill (2007, 2015)

In his paper [Gil15], Asif Qumer Gill analyzes six modeling standards in terms of their applicability in an agile enterprise architecture. He highlights the necessity of first

creating basic models in ArchiMate in an agile setting and then detailing them step by step. This approach serves to implement a top-down model. At the same time, the author points out that detailed models such as BPMN, Unified Modeling Language (UML) or Framework for Agent-Oriented Method Engineering Language (FAML) already exist in many organisations. Therefore, the integration of a relatively new modeling language such as ArchiMate requires a bottom-up approach.

ArchiMate is chosen as the modeling language for the enterprise architecture to evaluate the applicability of heterogeneous modeling languages. BPMN, UML, FAML, Service-Oriented Architecture Modeling Language (SoaML) and Business Motivation Model (BMM) are considered for the more detailed modeling language. The author evaluates six different modeling standards, each designed for different purposes. The evaluation is based on a two-dimensional table that analyzes which elements of the five modeling languages can be assigned to the Business, Application, Technology and Extension layer. The result is an overview that shows the relationships between ArchiMate elements and the corresponding elements from the other modeling languages. In summary, a hybrid approach is proposed in which modeling is done using ArchiMate and detailed models are developed step by step and iteratively as needed [Gil15].

The hybrid approach proposed in [Gil15] is examined in [GQ15] using an interoperability framework specifically for ArchiMate and BPMN. This research analyzes syntactic, semantic and structural perspectives and identifies their similarities. This is followed by an ontological analysis to critically evaluate the initially created list based on the meta-models of both languages. The result of this two-stage analysis is shown in Figure 3.8.

ArchiMate	BPMN
Business Actor	Participant
Business Role	Partner Role
Business Collaboration	Collaboration
Business Interface	Interface
Location	Pool
Business Event	Event
Data Object	Data Object
Business Service	Activity
Business Process	Activity
Artifact	Artifact

Figure 3.8: Candidate Concept between ArchiMate and BPMN. Adopted from [GQ15].

The analysis results show that the ontological-semantic analysis in [GQ15] led to a different mapping of element pairs between the two languages than the logical evaluation in [Gil15].

This new assignment of concepts serves as the basis for developing a merged meta-model. Although this model initially appears to contain more elements than the individual

meta-models of the respective languages or the combination of both [Gil15], a quality assessment in terms of size and complexity proves that this is not the case. [GQ15].

In their work [GQ15], Gill et al. make a valuable contribution to the analysis of the mapping of elements between ArchiMate and BPMN and also propose a hybrid meta-model. However, the question of filtering remains unanswered, as a concrete implementation of this hybrid meta-model has not yet been further researched and remains open.

However, it remains unclear how the merged metamodel can be used in practice, especially if only a few elements of both languages have been related to each other. This result is in line with expectations based on previous research, but it raises the question of how complete models can be created with this hybrid metamodel if only a subset of similarities has been identified. While both the top-down and bottom-up approaches are discussed in [Gil15], a detailed examination of these approaches is missing in [GQ15].

3.7 Comparison of Synchronization Approaches

This section provides a comparison of the approaches for synchronizing enterprise architecture and business process modeling that were summarized above. A key factor for comparison is the “Link Type” attribute, which was already presented in the section 3.1. This attribute differentiates the approaches according to whether they provide for a direct link between the elements or the introduction of a new meta model to intercede between the standards. In the case of the direct linking approaches, no difference was made between 1-to-1 or many-to-many relationships, as this is not of primary importance for this comparison. Another important criteria is the synchronization direction. This attribute indicates whether an approach provides for top-down, bottom-up or bidirectional synchronization. Both synchronization directions are relevant for this work, since a complete bidirectional synchronization is being evaluated. Nevertheless, approaches that only support one direction are also considered, since they also provide valuable scientific results.

Table 3.2 shows a comparative overview of the approaches analyzed in this chapter for synchronization between enterprise architecture and business process modeling.

- Lankhorst researches in [vLtDL04] the so-called Workbench idea, which uses an approach that enables bidirectional synchronization. He concludes that a new meta-model is better suited as an intermediary than a direct link between elements.
- Penicina also proposes the use of a new meta model in [Pen13] and supports bidirectional synchronization.
- Gill et al. follow a similar approach and also examine a bidirectional synchronization based on an intermediary meta-model in [Gil15] and [GQ15].

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Nr	Ref.	Author(s)	Link-Type	Year	Sync. Direction	Proven
1	[vLtDL04]	M. Lankhorst et al.	New meta model	2004	top-down bottom-up	no
2	[OK20]	D. Orlovsky et al.	Direct	2020	bottom-up	no
3	[Pen13]	L. Penicina	New meta model	2023	top-down bottom-up	yes
5	[GQ15]	A. Q. Gill	New meta model	2015	top-down bottom-up	yes

Table 3.2: Comparison of the approaches according to defined criteria.

- Orlovskiy and Kopp, on the other hand, take a different approach in [OK20]. They investigate a direct link between the elements without using an additional meta-model. Their work focuses on bottom-up synchronization.

This tabular comparison shows that most approaches favor a bidirectional synchronization via an additional meta-model, while only a few works investigate a direct linking of the elements.

A comparison of the approaches shows that, although Lankhorst in [vLtDL04] created the basis for synchronization, the work of Penicina in [Pen13] offers the most mature approach for bidirectional synchronization to date. In addition, this approach has been validated by case studies, which underlines its practical applicability.

Summary and revisit of the Research Questions

The synchronization of ArchiMate and BPMN represents a challenge in the modeling of enterprise architectures and business processes. This thesis addressed the questions “*How can ArchiMate and BPMN be linked in such a way that synchronization across different levels of abstraction is ensured without semantic deviations?*” and “*What explicit filter mechanisms do these synchronization strategies provide when generating ArchiMate models from BPMN models?*”

To answer this questions, a critical literature review was carried out, supplemented by a tabular comparison of the identified approaches. ArchiMate provides a structural and strategic perspective on enterprise architecture, while BPMN enables detailed process modeling with a high level of granularity of process flows. The conceptual differences between these two modeling standards require specific methods for consistent synchronization. In this work, different synchronization approaches were analyzed that differ in terms of methodology, technical implementation, and synchronization direction.

The results of the literature research have shown that synchronization can be carried out in two different ways: on the one hand by a direct semantic link between elements from ArchiMate and BPMN, on the other hand by using a new meta model that serves as a logical intermediary layer between these standards. When linking elements directly, either a 1-to-1 assignment or a many-to-many relationship can be created. However, the analysis has shown that a direct 1-to-1 link between the models is not possible in all cases due to semantic and structural differences. The practical implementation of the direct link also has limits because the models often semantically differ. The intersection of compatible elements from both standards is often too small, which makes synchronization considerably more difficult.

Lankhorst developed a method that enables bidirectional synchronization with the Workbench approach. His approach was based on the concept of viewpoints, which only allow a partial representation of certain model contents. Synchronization is done via an additional meta model. However, no concrete implementation of the “tool-specific adapter” was developed in this work, which means that a practical implementation remains to be done. Filtering could be developed in the integration tier in the workbench architecture.

Orlovskiy and Kopp propose an approach that provides a 1-to-1 mapping between ArchiMate and BPMN. They use RDF graphs that are subsequently processed with SPARQL queries. The mapping of elements is done using a modified Breadth-First Search Algorithm. A significant limitation of this approach is the incorrect interpretation of BPMN pools, which were erroneously classified as ArchiMate “Business Process” elements. In addition, this approach only supports bottom-up synchronization and therefore no filter mechanism is applicable.

Penicina suggests an ontological approach that uses the Bunge-Wand-Weber (BWW) model as a semantic basis. This approach classifies the elements of both modeling standards and identifies conceptual similarities. However, Penicina’s analysis also shows that there are discrepancies in the mapping, which can lead to gaps in synchronization. The new meta model recommended here can be used in the filtering of elements.

Gill proposes a high-level ArchiMate model that represents a combination of the two directions, top-down and bottom-up, and is iteratively transferred into detailed BPMN processes. The ontological analysis shows that different perspectives can lead to different mappings of the model elements. However, it remains unclear how a complete meta-model for synchronization can be implemented in practice. As there is no specific implementation proposal for this hybrid meta-model, filtering was not considered in this approach.

This critical literature review has shown that bidirectional synchronization between ArchiMate and BPMN is feasible in principle, but that it brings different challenges.

The meta-model-based approaches offers flexible and theoretically sound solutions, but have yet to be proven in practice. Direct mapping of the elements offers a simple way to link models, but has semantic inconsistencies.

In summary, a new meta-model as an intermediate layer between ArchiMate and BPMN provides an effective solution for synchronization. The introduction of unified concepts and rules could ensure consistent transformation between the two modeling standards. This approach also helps to minimize semantic differences by defining specific mapping rules for the transformation within the intermediate layer. The meta-model would serve as an additional level of abstraction and enable a unified view of both standards.

The second research question regarding filtering options could only be answered to a certain level. One of the reasons for this is that synchronization filtering must be based on an existing synchronization approach. However, as no comprehensive synchronization approach exists, it can be concluded that no specific filter mechanisms have been developed.

However, the literature research has shown that approaches with meta-models as an intermediate layer could offer potential opportunities for implementing filter mechanisms. Nevertheless, no specific filter solution was identified in any of the approaches examined.

A complete implementation of a synchronization of the two modeling standards ArchiMate and BPMN requires further research. In particular, there is still a lot to be explored in the area of filter options for controlling the transfer of information.

This part of the question remains unanswered, on the one hand, the research has not provided any results, and, on the other hand, filtering is possible based on a solution for synchronization.

Overview of Generative AI Tools Used

- The translation site **DeepL** (<https://www.deepl.com>) was used for Translations from German into English.
- **ChatGPT** (<https://chatgpt.com>) was used to help check the grammar of texts and for syntax help with LaTeX.

Übersicht verwendeter Hilfsmittel

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Bibliography

- [Buc09] *Model-View-Controller Pattern*, pages 353–402. Apress, Berkeley, CA, 2009.
- [Gil15] Asif Qumer Gill. Agile enterprise architecture modelling: Evaluating the applicability and integration of six modelling standards. *Inf. Softw. Technol.*, 67:196–206, 2015.
- [GLB19] Fernanda Gonzalez-Lopez and Guillermo Bustos. Integration of business process architectures within enterprise architecture approaches: A literature review. *Engineering Management Journal*, 31(2):127–140, 2019.
- [GQ15] Asif Qumer Gill and Muhammad Atif Qureshi. Adaptive enterprise architecture modelling. *J. Softw.*, 10(5):628–638, 2015.
- [Lan17] Marc M. Lankhorst, editor. *Enterprise Architecture at Work - Modelling, Communication and Analysis, Fourth Edition*. Springer, 2017.
- [OK20] Dmytro Orlovskyi and Andrii Kopp. Enterprise architecture modeling support based on data extraction from business process models. In Sergey Subbotin, editor, *Proceedings of The Third International Workshop on Computer Modeling and Intelligent Systems (CMIS-2020), Zaporizhzhia, Ukraine, April 27-May 1, 2020*, volume 2608 of *CEUR Workshop Proceedings*, pages 499–513. CEUR-WS.org, 2020.
- [Pen13] Ludmila Penicina. Linking bpmn, archimate, and BWW: perfect match for complete and lawful business process models? In Janis Grabis, Marite Kirikova, Jelena Zdravkovic, and Janis Stirna, editors, *Short Paper Proceedings of the 6th IFIP WG 8.1 Working Conference on the Practice of Enterprise Modeling (PoEM 2013), Riga, Latvia, November 6-7, 2013*, volume 1023 of *CEUR Workshop Proceedings*, pages 156–165. CEUR-WS.org, 2013.
- [PK13] Ludmila Penicina and Marite Kirikova. Towards completeness and lawfulness of business process models. In Andrzej Kobylinski and Andrzej Sobczak, editors, *Perspectives in Business Informatics Research - 12th*

International Conference, BIR 2013, Warsaw, Poland, September 23-25, 2013. Proceedings, volume 158 of *Lecture Notes in Business Information Processing*, pages 63–77. Springer, 2013.

- [SSA24] Aisha Edna Alicia Sanyoto, Mochamad Chandra Saputra, and Ismiarta Aknuranda. Analyzing the impact of uml, bpmn, and archimate integration from user perspective. In *Proceedings of the 2024 10th International Conference on Computing and Artificial Intelligence, ICCAI 2024, Bali Island, Indonesia, April 26-29, 2024*, pages 409–414. ACM, 2024.
- [TOG22] The Open Group. *ArchiMate 3.2 Specification*, 2022. Accessed: 2025-02-13.
- [vLtDL04] Diederik van Leeuwen, Hugo W. L. ter Doest, and Marc M. Lankhorst. A tool integration workbench for enterprise architecture - integrating heterogeneous models and tools. In *ICEIS 2004, Proceedings of the 6th International Conference on Enterprise Information Systems, Porto, Portugal, April 14-17, 2004*, pages 470–478, 2004.
- [vRWCdM15] Mark von Rosing, Stephen White, Fred Cummins, and Henk de Man. Business process model and notation - BPMN. In Mark von Rosing, Henrik von Scheel, and August-Wilhelm Scheer, editors, *The Complete Business Process Handbook: Body of Knowledge from Process Modeling to BPM, Volume I*, pages 429–453. Morgan Kaufmann/Elsevier, 2015.
- [WW93] Yair Wand and Ron Weber. On the ontological expressiveness of information systems analysis and design grammars. *Inf. Syst. J.*, 3(4):217–237, 1993.